

Brain Tumor Segmentation Using a 3D U-Net (BraTS2020)

A controlled ablation of normalization & augmentation strategies

Submitted by

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- Introduction & Motivation
- Problem Statement
- Objectives of the Project
- Literature Review / Existing Systems
- Proposed System / Approach
- Implementation Details
- Evaluation & Expected Results
- Justification
- Conclusion
- References

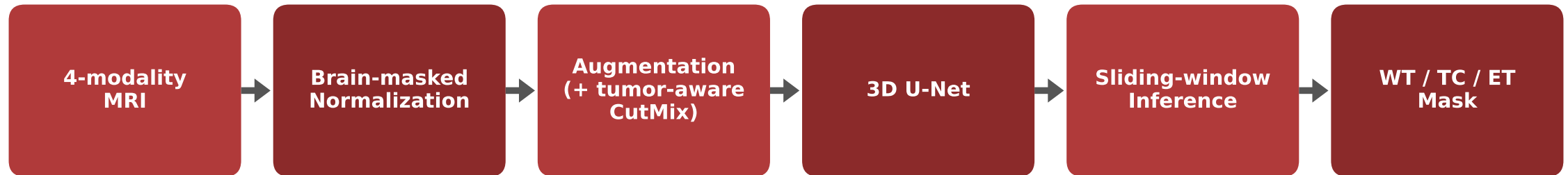
- Brain tumors (gliomas) are abnormal growths that disrupt neurological function and are life-threatening if not detected early.
- MRI is the standard imaging tool, but manual tumor segmentation is slow, expert-dependent, and varies between radiologists.
- Deep learning — especially the U-Net architecture — can automatically delineate tumor regions voxel-by-voxel in 3D.
- Motivation: build an accurate, reproducible 3D segmentation system and go further — systematically test which preprocessing and augmentation choices actually improve it, rather than assuming defaults.

- Tumors vary widely in size, shape, and location; small or irregular ones are easily missed.
- MRI intensities are not standardized — the same tissue looks different across scanners, so preprocessing choices strongly affect results.
- The tumor is under 5% of all voxels (severe class imbalance) — a naive model predicts 'all healthy' and still scores high on accuracy.
- Most projects fix one preprocessing recipe and never test alternatives — there is no controlled comparison of which choices help. This project addresses that gap.

- Develop an automated 3D U-Net system for multimodal MRI tumor segmentation on BraTS2020.
- Compare 4 intensity-normalization strategies (z-score, percentile-clip, WhiteStripe, hybrid) — ablation axis 1.
- Compare augmentation strategies, including an original tumor-aware CutMix targeting class imbalance — ablation axis 2.
- Evaluate with clinically standard metrics: Dice, Sensitivity, Specificity, and HD95, reported per region (WT / TC / ET).
- Ensure a leak-free, reproducible evaluation (patient-level split, fixed seeds).

- U-Net (Ronneberger, 2015) and 3D U-Net (Cicek, 2016) set the standard for medical segmentation via encoder-decoder + skip connections.
- V-Net (Milletari, 2016) introduced Dice loss to handle foreground/background imbalance in medical volumes.
- nnU-Net (Isensee, 2021) showed that careful configuration (preprocessing, augmentation) often matters more than architecture — motivating our ablation focus.
- CutMix (Yun, 2019) mixes image regions for regularization, but random box placement is ineffective when the target class is tiny — which our tumor-aware variant fixes.
- BraTS (Menze 2015; Baid 2021) is the benchmark; small-tumor detection and boundary precision remain open challenges.

- A PyTorch 3D U-Net segments tumors from the 4 co-registered MRI modalities (FLAIR, T1, T1ce, T2) stacked as input channels.
- MRI volumes are normalized over brain voxels only (one of 4 strategies under test) and augmented before training.
- The encoder-decoder with skip connections predicts a 4-class 3D mask (background + 3 tumor sub-regions).
- A Dice + Cross-Entropy loss combats class imbalance; full volumes are segmented via sliding-window inference.
- Two novel contributions: (1) a controlled normalization ablation, and (2) tumor-aware CutMix, which grafts a real tumor region into every mixed sample.



Patch-based 3D pipeline; full volumes reconstructed at inference by sliding-window stitching, then scored with Dice / HD95.

- Data: 369 BraTS2020 patients from NIfTI; 4 modalities stacked into a 4-channel 3D volume.
- Preprocessing: brain-masked intensity normalization (4 strategies compared).
- Split: patient-level train/validation split — prevents leakage between adjacent slices.
- Model: 3D U-Net with instance normalization (reliable at batch size 1-2) and skip connections.
- Training: patch-based (96^3), Dice+CE loss, mixed precision, gradient accumulation, cosine LR, early stopping.
- Augmentation: flips / rotations / elastic + gamma / bias-field / noise, plus tumor-aware CutMix.
- Inference: Gaussian-weighted sliding window over full volumes.

- After training, the model is tested on a held-out set of unseen patients.
- Predicted masks are compared to the dataset's ground-truth annotations.
- Metrics, reported per region (WT / TC / ET): Dice (overlap), Sensitivity (missed tumor), Specificity (false alarms), HD95 (boundary accuracy).
- Why not accuracy? With $<5\%$ tumor voxels, accuracy is misleading — Dice and HD95 are the meaningful measures.
- *Target from BraTS literature: WT Dice ~ 0.85 - 0.90 — actual results pending our experiments.*

Factor	Justification
Accuracy	3D U-Net captures volumetric tumor context for precise segmentation.
Automation	Removes slow, manual slice-by-slice annotation.
Early Detection	Reliable segmentation supports earlier treatment planning.
Efficiency	Segments a full volume in seconds vs. minutes manually.
Reproducibility	Fixed seeds + patient-level split + committed results ensure verifiable claims.
Rigor	Validated with Dice, Sensitivity, Specificity, HD95 — not just accuracy.

- We built a PyTorch 3D U-Net pipeline for BraTS2020 with a controlled two-axis ablation and an original tumor-aware CutMix augmentation.
- The framework produces leak-free, reproducible per-region metrics (Dice, Sensitivity, Specificity, HD95).
- Expected performance (from BraTS literature, to be confirmed): WT Dice $\sim$ 0.85-0.90, lower for the smaller TC / ET regions.
- **Key highlights:**
 - Modular, config-driven pipeline; every result traceable to a saved experiment file.
 - Novel tumor-aware CutMix directly targets the dataset's class imbalance.
 - Ablation will identify which normalization and augmentation choices genuinely help.
- *Actual Dice / HD95 numbers will be reported once the ablation runs complete.*

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